

AZURE CLOUD ENGINEERING

REAL-WORLD CAPSTONE DEPLOYMENT

Programme Purpose

The student has completed one month of Azure/AZ-900 fundamentals. This second month moves from learning Azure services to applying them as a Junior Cloud Engineer. Every session contributes to one realistic business project.

Client Scenario: Faktlens Digital Services

Faktlens Digital Services is a growing small business moving part of its IT environment to Microsoft Azure. The student is engaged as a Junior Cloud Engineer to design, deploy, secure, monitor, troubleshoot and document the solution.

Client Requirements

- Host the company website on Azure.
- Create a Resource Group and structured Azure environment.
- Build a VNet with appropriate subnets.
- Deploy a Windows Server VM for IIS.
- Configure Network Security Groups and secure administration.
- Create Azure Blob Storage for documents and images.
- Apply Microsoft Entra ID/RBAC concepts.
- Configure Azure Monitor, alerts and cost controls.
- Understand backup and disaster recovery.
- Evaluate App Service as a PaaS alternative.
- Introduce Azure Functions and Infrastructure as Code.
- Produce documentation and defend the solution.

Target Architecture

Internet → Azure VNet → WebSubnet → Windows VM/IIS → Website

↳ Azure Storage

ManagementSubnet → Administrative access

Azure Monitor → Metrics / Alerts / Activity Log

Naming Convention

Resource Group: Faktlens-Cloud-RG | VNet: Faktlens-VNet | Web subnet: WebSubnet | Management subnet: ManagementSubnet | VM: Faktlens-Web01 | Storage: globally unique faktlensstorageXXXX

Session 1: Azure Architecture & Solution Design

Duration: 2 hours

What the student must do

1. Read the client scenario and extract technical requirements.
2. Draw the target architecture showing Internet, VNet, both subnets, VM, NSG, Storage and Monitoring.
3. Choose an Azure region and record why.
4. Create a naming convention.
5. Explain why VM hosting is being used and identify a PaaS alternative.
6. Create a deployment checklist.

Required deliverables

- Architecture diagram
- Resource naming table
- One-page solution design
- Deployment checklist

Session 2: Azure Networking & Network Security

Duration: 2 hours

What the student must do

7. Create/open Faktlens-Cloud-RG.
8. Create Faktlens-VNet with 10.0.0.0/16.
9. Create WebSubnet 10.0.1.0/24.
10. Create ManagementSubnet 10.0.2.0/24.
11. Create/configure an NSG.
12. Plan HTTP 80, HTTPS 443 and restricted administrative access.
13. Associate the NSG appropriately.
14. Record every rule and its purpose.

Required deliverables

- VNet screenshot
- Subnet screenshots
- NSG rule table
- Public/private IP explanation

Session 3: Azure Virtual Machine Deployment

Duration: 2 hours

What the student must do

15. Create Faktlens-Web01.
16. Select an available Windows Server image.
17. Choose a small cost-conscious training VM size.
18. Place it in Faktlens-VNet/WebSubnet.
19. Configure public IP only where required.
20. Set administrator credentials securely; never put the password in the report.
21. Review inbound rules.
22. Deploy and record OS, size, region, subnet, IP and disks.

Required deliverables

- Running VM screenshot
- VM configuration record
- Network/public IP evidence

Session 4: IIS & Website Deployment

Duration: 2 hours

What the student must do

23. Connect to the VM using the approved administrative method.
24. Install Web Server (IIS) through Server Manager.
25. Confirm IIS is running.
26. Open C:\inetpub\wwwroot.
27. Copy the supplied index.html into the web root.
28. Test locally.
29. Test externally through the public IP.
30. If it fails, investigate IIS, Windows Firewall, NSG, public IP and port 80.
31. Document the final result.

Required deliverables

- IIS screenshot
- Live website screenshot
- Troubleshooting notes
- Website IP/URL

Session 5: Azure Storage & Blob Management

Duration: 2 hours

What the student must do

32. Create a globally unique Storage Account using Standard performance.
33. Choose a suitable training region and redundancy option.
34. Create containers: documents, images, backup.
35. Upload non-sensitive sample files.
36. Review access settings and avoid unnecessary public access.
37. Explore Hot/Cool/Archive concepts.
38. Explain how the business would use the storage.

Required deliverables

- Storage screenshot
- Container screenshots
- Test-file evidence
- Access/redundancy decision

Session 6: Microsoft Entra ID, RBAC & Security

Duration: 2 hours

What the student must do

39. Review available Entra ID tenant/users.
40. Create/use appropriate test identities where permitted.
41. Design access for CEO, IT Administrator and Finance Officer.
42. Identify suitable built-in RBAC roles.
43. Assign permissions at the narrowest sensible scope.
44. Review MFA concepts.
45. Document the design without passwords/secrets.

Required deliverables

- Role matrix
- RBAC evidence
- Security decisions

Session 7: Azure Monitor, Alerts & Cost Management

Duration: 2 hours

What the student must do

46. Open Azure Monitor and inspect VM metrics.
47. Review CPU, network and availability information where available.
48. Create a useful training alert.
49. Review Activity Log.
50. Review Resource Health.
51. Open Cost Management.
52. Create a budget/cost alert if permitted.
53. Write what you would do after an unexpected cost increase.

Required deliverables

- Monitoring screenshots
- Alert evidence
- Activity Log evidence
- Budget/cost evidence
- Incident response note

Session 8: Break/Fix Troubleshooting Lab

Duration: 2 hours

What the student must do

54. INC-001: diagnose and restore an unavailable website.
55. INC-002: diagnose inability to administer the VM.
56. INC-003: diagnose a storage access problem.
57. For each incident, check application, OS, network and identity layers as appropriate.
58. Record symptoms, evidence, root cause, fix and prevention.
59. Verify the service after each fix.

Required deliverables

- Three incident reports
- Before/after evidence
- Root-cause notes

Session 9: Azure SQL & Managed Database Concepts

Duration: 2 hours

What the student must do

60. Review the need for a future customer database.
61. Compare SQL Server on a VM with Azure SQL Database.
62. If permitted, deploy a small training Azure SQL resource.
63. Use only sample data.
64. Review access controls.
65. Explain why managed database services can reduce operational responsibility.
66. Delete/stop training resources when finished.

Required deliverables

- Architecture comparison
- Deployment evidence if used
- Cost/security recommendation

Session 10: Backup & Disaster Recovery

Duration: 2 hours

What the student must do

67. Define RPO and RTO for the website and documents.
68. Review Azure Backup/Recovery Services concepts.
69. If permitted, configure a safe training backup.
70. Identify what needs backup and how often.
71. Discuss VM failure, file deletion and regional outage scenarios.
72. Write a simple recovery procedure.
73. Identify limitations of the recovery plan.

Required deliverables

- RPO/RTO table
- Backup evidence if available
- Recovery procedure

Session 11: Azure App Service: PaaS Alternative

Duration: 2 hours

What the student must do

74. Review the VM/IIS architecture.
75. If permitted, create a small App Service.
76. Deploy the supplied website.
77. Test the App Service URL.
78. Compare VM and App Service responsibilities.
79. Recommend when the client should use each.
80. Clean up if not required.

Required deliverables

- Working App Service evidence if deployed
- IaaS/PaaS comparison
- Client recommendation

Session 12: Azure Functions & Serverless

Duration: 2 hours

What the student must do

81. Review a requirement to process an event automatically.
82. Explain VM vs event-driven serverless.
83. If permitted, create a basic Function App from a supported starter template.
84. Identify its trigger and output.
85. Test safely.
86. Document one business use case.
87. Delete training resources when finished.

Required deliverables

- Function architecture sketch
- Test evidence if deployed
- Business use case

Session 13: Infrastructure as Code with Bicep

Duration: 2 hours

What the student must do

88. Explain manual deployment vs IaC.
89. Open Cloud Shell or another suitable environment.
90. Create a simple Bicep template for a training resource such as storage or a VNet.
91. Use parameters.
92. Validate it.
93. Deploy it to the project Resource Group.
94. Verify the resource.
95. Document the benefit of repeatability.

Required deliverables

- Bicep file
- Deployment evidence
- IaC explanation

Session 14: Cloud Security Hardening Challenge

Duration: 2 hours

What the student must do

96. Review the environment without changing it initially.
97. Identify unnecessary public exposure.
98. Review and reduce NSG access.
99. Review administrative access.
100. Review storage exposure.
101. Review excessive RBAC assignments.
102. Check monitoring/alerts.
103. Create prioritized security recommendations.
104. Implement safe improvements and retest.

Required deliverables

- Security assessment
- Before/after evidence
- Prioritized remediation list

Session 15: Final Capstone: Build the Client Environment

Duration: 2 hours

What the student must do

105. Re-read requirements and create final architecture.
106. Verify Resource Group.
107. Deploy VNet/subnets.
108. Deploy VM.
109. Configure NSG and secure administration.
110. Install IIS and deploy website.
111. Create/configure Blob Storage.
112. Apply identity/RBAC controls.
113. Configure monitoring and an alert.
114. Configure budget/cost alert where available.
115. Document backup/DR considerations.
116. Test services.
117. Produce final technical report and prepare a 5–10 minute presentation.

Required deliverables

- Working Azure environment
- Architecture diagram
- Technical report
- Testing evidence
- Troubleshooting log
- Security review
- Presentation

Session 16: Project Defence & Professional Review

Duration: 2 hours

What the student must do

118. Present the architecture.
119. Demonstrate the live website.
120. Demonstrate storage.
121. Explain network/security configuration.
122. Show monitoring and cost controls.
123. Explain one troubleshooting incident.
124. Answer architecture, security, cost and scalability questions.
125. Identify three production improvements.
126. Submit final report.

Required deliverables

- Presentation
- Final report
- Working demonstration
- Self-review

Final Assessment Rubric

- Architecture & design — 15 marks
- Azure deployment — 20 marks
- Networking & security — 15 marks
- Website/application deployment — 10 marks
- Monitoring & cost management — 10 marks
- Troubleshooting — 10 marks
- Documentation — 10 marks
- Presentation & professional communication — 10 marks

Student Submission Checklist

- Architecture diagram
- Azure resource inventory
- Network/NSG configuration
- VM configuration
- Website test
- Storage configuration
- RBAC/identity design
- Monitoring/alert evidence
- Cost-management evidence
- Backup/DR plan
- Troubleshooting reports
- Security assessment
- Final technical report
- Final presentation

Azure Cost & Safety Rules

- Use training/test resources only.
- Use small, cost-conscious resource sizes that are actually available.
- Check current pricing and free-service eligibility in the student's subscription.
- Stop/delete chargeable resources when no longer needed.
- Never use real customer credentials or confidential data.
- Never put passwords, keys or secrets in screenshots or reports.
- If a feature is unavailable because of subscription/tenant restrictions, document the limitation instead of inventing a result.